

The CFFM-Net Pilot Study

Problem, novelty and position in the literature

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October 2026

ABSTRACT. The pilot builds and evaluates the still-image form of CFFM-Net: a dual-stream YOLO26 detector whose visible and thermal features are fused by a selective state-space scan. In that scan, the discretisation step of every token is scaled by an estimate of how reliable its modality is at its location. This note states the problem precisely and identifies twelve recent papers whose limitations bear on it. It separates the limitations their authors state from those observed in their reported results. It then specifies five contributions of the pilot, each tied to at least one of those limitations and to an experiment that can refute it. The pilot addresses six stated limitations from five papers. Two further stated limitations, temporal memory and post-export accuracy, are deferred to later phases and listed as such. Claims are framed as hypotheses. A two-day pilot with one training seed per configuration can support or undermine them, but cannot establish them.

1 The problem

Detecting very small objects with visible and thermal cameras is the regime in which each sensor fails differently. Visible imagery loses objects in darkness, glare and fog. Thermal imagery loses texture and confuses objects with backgrounds at the same temperature. Both lose objects that cover only a few pixels. Three properties are therefore needed together:

- enough spatial resolution to see objects of a few pixels;
- a fusion rule that trusts each sensor only where it is informative;
- a cost low enough for real-time use.

State-space models (SSMs) are attractive because they mix information globally at linear cost [12, 19]. The recent literature has shown, however, that dropping a Mamba block into a detector is

neither cheap nor clearly useful (Section 2). The pilot asks a narrower question: *can a selective scan be made to fuse two sensors selectively, at the resolution where small objects live, at a cost compatible with a real-time YOLO detector?*

2 What the latest work leaves open

Table 1 lists twelve recent papers. For each it gives one limitation and how that limitation is known. *Stated* means the authors write it themselves. *Measured* means it follows from numbers the authors report. *Inferred* means it follows from the method as published. The last column gives the contribution of Section 3 that addresses the limitation, or the later phase to which it is deferred.

Two patterns stand out. First, the cost of SSM blocks is the most frequently reported weakness. It is measured in CFMW, stated in UAVDet and stated in general terms by SF-Mamba. Second, small and tiny objects remain the acknowledged failure case of the best recent detectors, with or without an SSM. Fusion work also assumes, almost everywhere, that both sensors are equally trustworthy at every pixel. The fusion-weighting schemes of CFMW and Fusion-Mamba act on the output of the scan, after a degraded modality has already written into the state.

3 Contributions of the pilot

Figure 1 connects the stated and measured limitations to the five contributions. Each contribution is described with its closest prior work and the pilot experiment that tests it.

C1. Reliability-gated cross-modal selective scan

WHAT IT IS. At every location, a small head estimates a reliability $r^m \in (0, 1)$ for each modality m from the two feature maps and their absolute difference. The visible and thermal tokens of each location are *interleaved* in one sequence, so the hidden state alternates between modalities at every position. The selective step of token k is then scaled by the reliability of its modality at its location:

$$\Delta_k = \text{softplus}(w^\top x_k + b) \cdot r_{p(k)}^{m(k)}. \quad (1)$$

Because $\bar{A}_k = \exp(\Delta_k A)$ and $\bar{B}_k \approx \Delta_k B_k$, a token judged unreliable ($r \rightarrow 0$) neither writes into the state nor erases it [12]. A displacement field resamples thermal features onto the visible grid before the scan, following COMO [4]. A depthwise convolutional branch beside the scan restores local two-dimensional structure.

TABLE 1. Limitations of recent work and how the pilot responds. C1 to C5 are the contributions of Section 3.

Paper, venue, date	Limitation	Basis	Response
CFMW, IEEE TCSVT, Dec 2025 [18]	still misses small objects, which are 26% of its benchmark	stated	C2
	the scan triples computation (95.6 to 308.6 GFLOPs) for +3.2 mAP; 13.5 FPS	measured	C3
UAVDet, CVIU, Jan 2026 [42]	fails in very dark scenes; not validated on multimodal input; multimodal and low-light use named as future work	stated	C1
	the sequential state update uses GPU parallelism poorly; adding Mamba costs 23 FPS for +0.7 AP	stated, measured	C3
HPS-DETR, IEEE TGRS, Jun 2025 [38]	objects below 10×10 pixels and distant objects are still missed, especially in snow	stated	C2
SF-Mamba, Mar 2026 [33]	Mamba is relatively slow at the short token lengths typical of vision	stated	C3
C-GSPN, May 2026 [2]	SSMs and linear attention weaken two-dimensional spatial structure	stated	C1
MambaOut, CVPR 2025 [46]	whether the SSM is needed for detection is left open	stated	C4
MambaVision, CVPR 2025 [14]	SSM and attention run only at strides 16 and 32; no small-object metric; no real-time detector	inferred	C2
Fusion-Mamba, IEEE TMM 2025 [9]	287.6M parameters and 78 ms per image; fusion assumes registered pairs	measured, inferred	C1, C3
Mamba YOLO, AAAI 2025 [39]	large model below YOLO11-l (52.1 against 53.4 AP); latency on an RTX 4090, not comparable with T4 figures	measured	C5
RGBT-Tiny, IEEE TPAMI 2025 [43]	IoU-based evaluation is unfair to tiny objects	stated	C2, C5
MambaST, ITSC 2024 [22]	fuses only three frames; not evaluated on longer sequences or other datasets	stated	Phase 2
LCMamNet, Jul 2026 [16]	accuracy falls after TensorRT FP16 export (71.27 to 65.96 mIoU)	measured	Phase 3

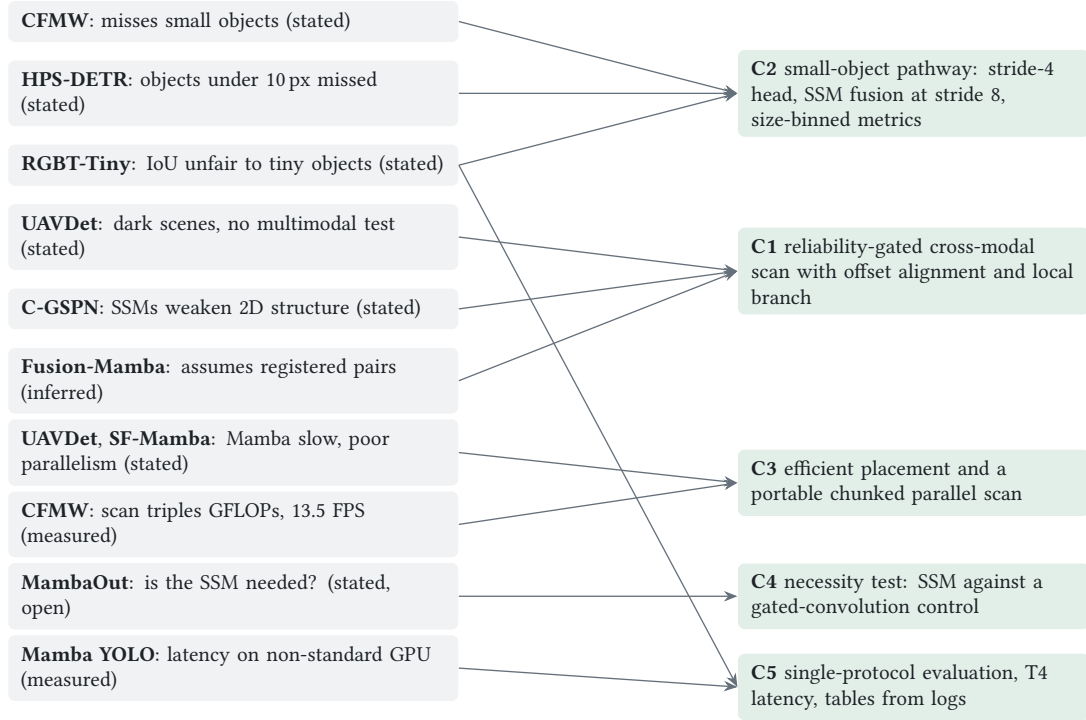


FIGURE 1. Limitations reported in recent work (left) and the pilot contribution that addresses each (right). Every limitation on the left is stated by its authors or follows from numbers they report, except the registration assumption, which follows from the published method.

WHY IT IS NEW. CFMW and Fusion-Mamba weight the modalities on the *output* of the scan [9, 18]. MambaRefine-YOLO gates by global illumination and modality difference [21]. Gating the discretisation step per location and per modality controls what enters the state itself. No published fusion method found in the survey does this.

LIMITATIONS ADDRESSED. UAVDet’s failure in dark scenes and its untested multimodal setting. C-GSPN’s loss of two-dimensional structure, through interleaving, which keeps the two modalities of each location adjacent, and through the local branch. Fusion-Mamba’s registration assumption, through offset alignment.

PILOT TEST. Three comparisons, all under identical training:

- CFFM-Net against the same network with $r \equiv 1$ (gating off);
- both against a plain two-stream concatenation baseline;
- a test-time probe that degrades one modality and shifts the thermal image, which requires no extra training.

C2. A small-object pathway in a real-time dual-stream YOLO

WHAT IT IS. A stride-4 detection head is grafted onto YOLO26 [45]. The SSM fusion is applied from stride 8, not only at strides 16 and 32 as in MambaVision [14]. Evaluation reports AP in the AI-TOD size bands (2–8, 8–16 and 16–32 pixels) alongside COCO AP_S.

LIMITATIONS ADDRESSED. Small objects missed by CFMW. Objects under ten pixels missed by HPS-DETR. RGBT-Tiny’s warning about IoU-based evaluation, through size-binned reporting now and SAFit once RGBT-Tiny access is granted.

PILOT TEST. Size-binned AP of CFM-Net against the two-stream baseline with and without the stride-4 head.

C3. Efficient placement and a portable parallel scan

WHAT IT IS. Two decisions keep the cost of the scan down.

- Scans run where sequences are long. Interleaving doubles the sequence length, to 10 240 tokens at stride 8 for a 640×512 input. No scan runs at stride 4, where a reliability-weighted sum suffices.
- The selective scan is implemented twice and tested for numerical agreement: with the official CUDA kernel where available, and as a two-level chunked parallel scan in plain PyTorch. The PyTorch version needs about $2\sqrt{L}$ sequential steps instead of L , and it runs on Windows laptops and on GPUs that the CUDA kernel does not support.

LIMITATIONS ADDRESSED. UAVDet’s poor GPU parallelism, SF-Mamba’s slow short sequences, and the measured cost of CFMW and Fusion-Mamba.

PILOT TEST. Parameters, GFLOPs and batch-one latency on a T4 for every model, including NMS where it is used.

C4. A necessity test for the SSM

WHAT IT IS. A control model replaces the selective scan in CMFM with a gated depthwise convolution of matched cost. Everything else, including reliability gating of the input, is kept.

LIMITATION ADDRESSED. MambaOut found the SSM unnecessary for ImageNet classification and left detection open [46]. The analysis of Mamba as gated linear attention [13] makes the same control necessary.

PILOT TEST. CFFM-Net against the control on LLVIP. A null result is reported, not hidden.

C5. A single measurement protocol

WHAT IT IS. All of these run through one code path:

- every baseline is retrained, not copied from a paper;
- one evaluation script scores every model;
- latency is measured on a T4, the GPU used by the YOLO and DETR literature;
- every table is generated from logged runs.

LIMITATION ADDRESSED. The incomparable latency figures of Mamba YOLO. The fifteen internal inconsistencies found in the four source papers of the project.

4 Scope of the pilot

WHAT THE PILOT DOES. It trains nine configurations at the smallest YOLO26 scale with a short schedule and one seed each, on LLVIP and M3FD, the two freely downloadable paired benchmarks (Table 2). The planned total is about 26 T4 GPU-hours, run in parallel on Kaggle T4s and a local RTX 4050.

TABLE 2. Pilot configurations. T4-hours are planning estimates.

Configuration	Data	Tests	T4-hours
YOLO26-n, visible only	LLVIP	baseline	1.5
YOLO26-n, thermal only	LLVIP	baseline	1.5
two-stream YOLO26-n, concatenation	LLVIP	baseline	2.5
CFFM-Net (image mode)	LLVIP	C1–C3	4.0
CFFM-Net, gating off ($r \equiv 1$)	LLVIP	C1	4.0
CFFM-Net, scan replaced by gated convolution	LLVIP	C4	4.0
two-stream YOLO26-n, concatenation	M3FD	baseline	1.5
CFFM-Net (image mode)	M3FD	C1–C3	2.0
test-time degradation probe	LLVIP	C1	0.5
environment checks, latency (C5)		C3, C5	4.5
<i>total</i>			<i>about 26</i>

WHAT THE PILOT DOES NOT CLAIM.

- Temporal memory across frames. That is CFFM-Net-S, which answers MambaST’s stated limitation in Phase 2.
- Robustness to sensor dropout over time, and the full robustness protocol.
- Accuracy after TensorRT export on embedded hardware, which answers LCMamNet’s observation in Phase 3.
- Results on RGBT-Tiny, which await dataset approval.
- Statistical significance. One seed per configuration cannot separate small effects from run-to-run variation, which HPS-DETR’s own tables suggest is about 1 to 1.5 points [38].

5 Pilot hypotheses

The pilot is designed so that each hypothesis can come out against the method (Table 3). The thresholds exceed the single-seed noise estimated above wherever possible. A result inside that noise is reported as inconclusive.

TABLE 3. Pilot hypotheses and decision rules. AP is COCO AP_{50:95}.

	Hypothesis	Supported if
PH1	Fusion beats the best single sensor at night	CFFM-Net exceeds thermal-only YOLO26-n on LLVIP by at least 1.5 AP
PH2	Gating the step adds value	CFFM-Net exceeds the gating-off model by at least 1.5 AP, or degrades less in the test-time probe by the same margin
PH3	The pathway helps small objects	AP in the 8–32 pixel bands exceeds the two-stream baseline on M3FD by at least 1.5
PH4	The SSM is needed	CFFM-Net exceeds the gated-convolution control by at least 1.0 AP at matched GFLOPs
PH5	The cost is acceptable	T4 latency of CFFM-Net is no more than twice that of the two-stream baseline

6 Related work

The full survey, newest first, is Chapter 4 of the research dossier. This section places the pilot within it.

REAL-TIME DETECTION. The accuracy–latency frontier in 2025–2026 belongs to two lines of work. YOLO26 removes NMS and adds a small-target-aware label assignment [45]. Detection transformers draw on DINOv2 and DINOv3 features: RF-DETR [29], RT-DETRv4 [30] and DEIMv2 [7]. YOLOv12 and YOLOv13 bring attention and hypergraph correlation into the YOLO family [17, 35].

The pilot builds on YOLO26 because its assignment already targets small objects and its modules are open to extension.

VISION STATE-SPACE MODELS. Several 2025–2026 backbones make SSMs competitive as general backbones: MambaVision [14], Spatial-Mamba [41], DAMamba [5] and PRISMamba [26]. The critical literature is as important.

- MambaOut questions whether the SSM is needed at all [46].
- MLLA recasts Mamba as gated linear attention [13].
- SF-Mamba and C-GSPN report cost and spatial-structure problems [2, 33].
- VNCT and GraphScan contest the need for directional scans [11, 37].

C3 and C4 respond directly to this critical line.

MAMBA INSIDE YOLO. Mamba YOLO established the Ultralytics-based baseline [39]. MambaPSA places Mamba inside YOLO26 and reports changes of -0.1 to $+0.9$ AP [20]. ScopeMamba-YOLO, YOLO12-MambaScan and HDMamba-YOLO target UAV small objects [15, 31, 44]. UAVDet shows that most of its gain comes from the stride-4 head and early capacity, not the Mamba block [42]. The pilot therefore does not claim a Mamba-in-YOLO block as its contribution. The block exists to perform gated fusion.

SMALL AND TINY OBJECTS. SFDNet, Dome-DETR and EFSI-DETR define the current state of the art [8, 10, 32]. RGBT-Tiny contributes the benchmark that matches the full project and a matching measure for tiny boxes [43]. HPS-DETR illustrates the value of fusing the stride-4 level [38].

VISIBLE–THERMAL FUSION. Fusion moved from attention, in CFT and ICAFusion [27, 34], to state-space blocks: Fusion-Mamba [9], CFMW [18], COMO [4], WaveMamba [40], MS2Fusion [23] and MambaRefine-YOLO [21]. In 2026, foundation-model fusion followed, with RegisterBridgeMM and CFGPNet [1, 28]. Misalignment is handled by offset guidance in COMO and by contextually guided SSM fusion [3, 4]. None gates the state update per modality and location, which is the step C1 adds.

TEMPORAL AND STREAMING DETECTION. MambaST fuses three visible–thermal frames [22]. TMambaDet runs a bidirectional temporal Mamba offline [36]. Zubić and colleagues detect with recurrent SSMs on event cameras [47]. Alignment-free visible–thermal video detectors use windows, graphs and hypergraphs without SSMs [6, 24, 25]. These define Phase 2, for which the pilot builds and tests the shared components.

7 Threats to validity

- *One seed.* Differences below about 1.5 AP are inconclusive and are reported as such.
- *LLVIP is close to saturated* for AP₅₀. The pilot reports AP_{50:95} and size bands, and adds M3FD.
- *Both pilot datasets are registered.* Misalignment is therefore tested only by synthetic shifts at test time. Real misalignment waits for UVT-VOD2024 and DroneVehicle.
- *Short schedules at the smallest scale* may favour simpler models. Rankings are confirmed in the full plan at the small (s) scale with three seeds.
- *Re-implemented baselines* may differ from the originals. Every difference is documented in the code.

References

- [1] CFGPNet. Visible–infrared object detection. *arXiv preprint* arXiv:2608.06205, 2026.
- [2] C-GSPN. Scaling parallel sequence models for vision. *arXiv preprint* arXiv:2606.00746, 2026.
- [3] Contextually-guided state space fusion for misaligned multispectral object detection. In *Proc. ACM Int. Conf. Multimedia (ACM MM)*, 2025. doi:10.1145/3746027.3754550.
- [4] COMO: Cross-Mamba interaction and offset-guided fusion for multimodal object detection. *Information Fusion*, vol. 125, 2026. *arXiv preprint* arXiv:2412.18076, 2024.
- [5] DAMamba: Vision state space model with dynamic adaptive scan. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. arXiv:2502.12627.
- [6] DCHNet and the DVT-VOD1000 benchmark for unaligned visible–thermal video object detection. *arXiv preprint* arXiv:2607.08191, 2026.
- [7] DEIMv2: Real-time object detection meets DINOv3. *arXiv preprint* arXiv:2509.20787, 2025.
- [8] Dome-DETR. Tiny object detection with density-oriented feature-query manipulation. In *Proc. ACM MM*, 2025. arXiv:2505.05741.
- [9] W. Dong et al. Fusion-Mamba for cross-modality object detection. *IEEE Trans. Multimedia*, vol. 27, 2025. arXiv:2404.09146.
- [10] EFSI-DETR. Real-time small-object detection transformer for UAV imagery. *arXiv preprint* arXiv:2601.18597, 2026.
- [11] GraphScan. Graph-based scanning for vision state space models. *arXiv preprint* arXiv:2605.11300, 2026.
- [12] A. Gu and T. Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint* arXiv:2312.00752, 2023.
- [13] D. Han et al. Demystify Mamba in vision: A linear attention perspective. In *Advances in NeurIPS*, 2024. arXiv:2405.16605.

- [14] A. Hatamizadeh and J. Kautz. MambaVision: A hybrid Mamba–Transformer vision backbone. In *Proc. CVPR*, 2025. arXiv:2407.08083.
- [15] HDMamba-YOLO. Stage-wise mixed SSM–CNN detector for small objects. *arXiv preprint* arXiv:2609.23061, 2026.
- [16] LCMamNet. Lightweight Mamba network for infrared small targets, deployed on Jetson Orin NX. *arXiv preprint* arXiv:2607.24184, 2026.
- [17] M. Lei et al. YOLOv13: Real-time object detection with hypergraph-enhanced adaptive visual perception. *arXiv preprint* arXiv:2506.17733, 2025.
- [18] H. Li, Q. Hu, B. Zhou, Y. Yao, J. Lin, K. Yang, and P. Chen. CFMW: Cross-modality fusion Mamba for robust object detection under adverse weather. *IEEE Trans. Circuits and Systems for Video Technology*, 35(12):12066–12081, 2025. doi:10.1109/TCSVT.2025.3587918.
- [19] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, and Y. Liu. VMamba: Visual state space model. In *Advances in NeurIPS*, 2024. arXiv:2401.10166.
- [20] MambaPSA: A Mamba replacement for C2PSA in YOLO26. *arXiv preprint* arXiv:2607.12681, 2026.
- [21] MambaRefine-YOLO. Dual-gated Mamba fusion for visible–infrared small-object detection. *arXiv preprint* arXiv:2511.19134, 2025.
- [22] MambaST: A plug-and-play cross-spectral spatial-temporal fuser for efficient pedestrian detection. In *Proc. IEEE Int. Conf. Intelligent Transportation Systems (ITSC)*, 2024. arXiv:2408.01037.
- [23] MS2Fusion. Shared- and cross-parameter state-space fusion for multispectral object detection. *Information Fusion*, 2025. arXiv:2507.14643.
- [24] MSENNet and the UVT-VOD2024 benchmark for alignment-free visible–thermal video object detection. *arXiv preprint* arXiv:2410.12143, 2024.
- [25] MSGNet. Spatio-temporal graph network for alignment-free visible–thermal video object detection. *arXiv preprint* arXiv:2504.11779, 2025.
- [26] PRISMamba. Partial ring scan for vision state space models. In *Proc. ICML*, 2026. arXiv:2602.04170.
- [27] F. Qingyun, H. Dapeng, and W. Zhaokui. Cross-modality fusion transformer for multispectral object detection. *arXiv preprint* arXiv:2111.00273, 2021.
- [28] RegisterBridgeMM. Visible–infrared detection with a frozen DINOv3 encoder. *arXiv preprint* arXiv:2608.04833, 2026.
- [29] I. Robinson et al. RF-DETR: Neural architecture search for real-time detection transformers. In *Proc. ICLR*, 2026. arXiv:2511.09554.
- [30] RT-DETRv4. Real-time detection transformer distilled from DINOv3. *arXiv preprint* arXiv:2510.25257, 2025.
- [31] ScopeMamba-YOLO. Off-path selective scanning for small-object detection. *arXiv preprint* arXiv:2609.10156, 2026.
- [32] SFDNet. Spectrum-aware tiny-object detection. In *Proc. ECCV*, 2026. arXiv:2606.29029.
- [33] SF-Mamba. Efficient vision Mamba at short token lengths. *arXiv preprint* arXiv:2603.16423, 2026.
- [34] J. Shen et al. ICAFusion: Iterative cross-attention guided feature fusion for multispectral object detection. *Pattern Recognition*, 2024. arXiv:2308.07504.

- [35] Y. Tian, Q. Ye, and D. Doermann. YOLOv12: Attention-centric real-time object detectors. In *Advances in NeurIPS*, 2025. arXiv:2502.12524.
- [36] TMambaDet. Temporal Mamba for video object detection. In *Proc. CVPR*, 2026.
- [37] VNCT: Vision non-causal trapezoidal Mamba. *arXiv preprint* arXiv:2607.03589, 2026.
- [38] X. Wang and H. Chen. HPS-DETR: Enhancing small object detection with lightweight feature extraction and transformer integration. *IEEE Trans. Geoscience and Remote Sensing*, 63:5937420, 2025. doi:10.1109/TGRS.2025.3583418.
- [39] Z. Wang et al. Mamba YOLO: A simple baseline for object detection with state space model. In *Proc. AAAI Conf. Artificial Intelligence*, 2025. arXiv:2406.05835.
- [40] WaveMamba: Wavelet-driven Mamba fusion for RGB-infrared object detection. In *Proc. ICCV*, 2025. arXiv:2507.18173.
- [41] C. Xiao et al. Spatial-Mamba: Effective visual state space models via structure-aware state fusion. In *Proc. ICLR*, 2025. arXiv:2410.15091.
- [42] Y. Yang, F. Guo, and P. Niu. UAVDet: A CNN-Mamba hybrid network for efficient small object detection in UAV imagery. *Computer Vision and Image Understanding*, 264:104637, 2026. doi:10.1016/j.cviu.2026.104637.
- [43] X. Ying et al. Visible-thermal tiny object detection: A benchmark dataset and baselines. *IEEE Trans. Pattern Analysis and Machine Intelligence*, 2025. arXiv:2406.14482.
- [44] YOLO12-MambaScan. YOLO12 with high-frequency convolutions and Mamba context. *arXiv preprint* arXiv:2609.13647, 2026.
- [45] Ultralytics. YOLO26. Released January 2026; technical report *arXiv preprint* arXiv:2606.03748, 2026. <https://docs.ultralytics.com/models/yolo26>.
- [46] W. Yu and X. Wang. MambaOut: Do we really need Mamba for vision? In *Proc. CVPR*, 2025. arXiv:2405.07992.
- [47] N. Zubić, M. Gehrig, and D. Scaramuzza. State space models for event cameras. In *Proc. CVPR*, 2024. arXiv:2402.15584.